

Local Factor Graph Debate

A mathematical companion

Codex, Claude

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All text in this document is AI-generated. For a human-vetted version of this material, see the original LessWrong post.

Abstract

Two debaters know a graphical model that a judge can inspect only a little at a time. They take turns choosing variables to reveal; the judge then computes the target log-odds in the revealed submodel. How close is the result to the answer from the whole model? A small set of arguments first gives a strategic guarantee. Message passing then bounds the influence of everything outside that set. Gaussian averaging turns this bound into a power of the inspection budget. Finally, a projection argument on regular trees explains why a polynomial obstruction can remain even under optimal play. The guarantees concern agreement with the full model, and place no bound on the computation used by the judge or the players.

Acknowledgments. Stephan Wäldchen, Dmitry Vaintrob, and other collaborators contributed ideas to this project. Codex and Claude were used to generate and develop proofs; this companion was prepared with Codex. Mathematical arguments and numerical evaluations are distinguished below. No claim of complete machine verification is made for this document.

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1 The question: how much must the judge see?

Imagine a question surrounded by a large network of related claims. Some claims bear directly on the question; others matter only because they change how we interpret those nearby. Two informed advocates can each draw the judge’s attention to a few claims. If one argues for a high answer and the other for a low answer, can either systematically exploit what remains unseen?

The useful object will be a *core*: a small collection of claims that makes the answer stable under every further revelation. If either advocate can insist on showing that collection, both acquire a way to constrain the outcome. The equilibrium transcript need not contain it. A credible alternative strategy already constrains a zero-sum game’s value.

The rest of the argument asks when such a core exists. Along a chain of weak interactions, the effect of a distant claim is repeatedly diminished. Branching works in the opposite direction: there are more distant claims. On a graph with maximum degree d , a radius of R can cost on the order of $(d - 1)^R$ vertices. Thus exponential decay in distance becomes polynomial decay in the number of revelations. This change of scale is the geometric reason for the powers appearing later.

A random interaction table produces both an interaction and random local fields, as (7) makes explicit. A strong local field reduces the slope of the edge response (9), so increasing the scale of the tables need not make debate harder; Section 5 makes this quantitative at degrees three and four. Completion stability is also not the same as weak spin correlation: with zero fields and pure interactions $J_e x_u x_v / 2$, global spin-flip symmetry gives $L_S = 0$ for every induced submodel. All core oscillations then vanish, even when strong positive interactions make spins highly correlated.

For debate as a method of supervision, see [1, 2].

2 Problem formulation

2.1 A revealed claim adds a variable

Let $G = (V, E)$ be a finite simple undirected graph and $t \in V$ its target. A variable $x_v \in \{-1, +1\}$ represents whether claim v is false or true. A vertex has a unary *log-factor* $\varphi_v : \{-1, +1\} \rightarrow \mathbb{R}$. An edge $\{u, v\}$ has a binary log-factor $\theta_{(u,v)} : \{-1, +1\}^2 \rightarrow \mathbb{R}$, with

$$\theta_{(v,u)}(b, a) = \theta_{(u,v)}(a, b).$$

The reverse orientation names the same table. Each undirected edge is counted once. Missing unary factors are represented by $\varphi_v = 0$. All factors are finite; very large unary contrasts can approximate hard constraints, but infinite log-odds are outside our statements.

For a transcript $t \in S \subseteq V$, the judge uses

$$P_S(x) = \frac{1}{Z_S} \exp \left\{ \sum_{\substack{\{u,v\} \in E \\ u,v \in S}} \theta_{(u,v)}(x_u, x_v) + \sum_{v \in S} \varphi_v(x_v) \right\}, \quad x \in \{-1, +1\}^S. \quad (1)$$

Here Z_S is the sum of the numerator over all assignments, so P_S is a probability distribution. Write

$$L_S = \log \frac{P_S(x_t = +1)}{P_S(x_t = -1)}. \quad (2)$$

The reference answer is L_V , the log-odds from the entire model.

Revealing v exposes its unary factor and all edge tables joining it to already revealed vertices. It does not announce a value of x_v . The judge's distribution is the *induced submodel*: factors touching unrevealed vertices are omitted. It is generally different from the marginal of P_V on S . For example, with energy $Jx_t x_u / 2 + \ell x_u / 2$, deleting u gives target log-odds zero, whereas summing over x_u gives

$$\log \cosh \frac{\ell + J}{2} - \log \cosh \frac{\ell - J}{2},$$

which need not vanish. Selection itself is not treated as Bayesian evidence. These are substantive assumptions about the judge.

2.2 The game and its error

Both players know all tables and unary factors. Starting from $S = \{t\}$, Max and Min alternate revealing a previously unrevealed non-target vertex, with h moves each. Max wants the final L_S large and Min wants it small. Assume $2h \leq |V| - 1$, so every move is legal. For Max first, backward induction gives

$$D_{S,0} = L_S, \quad D_{S,\ell} = \begin{cases} \max_{v \notin S} D_{S \cup \{v\}, \ell-1}, & \ell > 0 \text{ even}, \\ \min_{v \notin S} D_{S \cup \{v\}, \ell-1}, & \ell \text{ odd}, \end{cases} \quad D_{2h} = D_{\{t\}, 2h}. \quad (3)$$

All our bounds also hold with Min first. The outcome is the value under optimal play. They are not guarantees for arbitrary player strategies. There is no penalty for computation and no option to change a revealed factor. A disconnected component of $G[S]$ cancels from the ratio in (2); only the component containing t affects the judge.

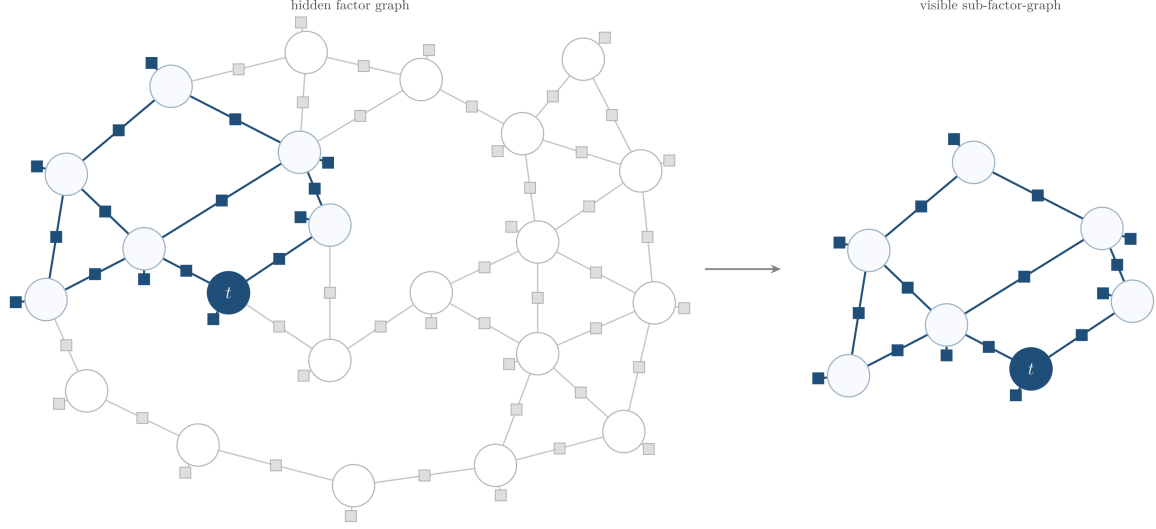


Figure 1: The hidden model and an induced transcript. Circles are variables; squares are factors. A factor survives precisely when every variable it touches has been revealed. The target t is present from the beginning.

We measure error in natural-log odds, $|D_{2h} - L_V|$. If the desired metric is probability, the sigmoid $s(x) = (1 + e^{-x})^{-1}$ satisfies

$$|s(x) - s(y)| \leq \tanh \frac{|x - y|}{4} \leq \frac{|x - y|}{4}. \quad (4)$$

For a fixed separation, its maximum occurs when x and y are symmetric about zero, which proves the first inequality. Thus a logit error of 0.1 implies a probability error below 0.025. A large logit error need not imply a large probability error when both logits are saturated.

There is no ground-truth assignment in the performance criterion. Agreement with L_V is agreement with the fully informed judge's model.

3 Cores and oscillation

3.1 A set that settles the question

A *core* is a connected set $K \subseteq V$ containing t . Its completion interval and its *oscillation* are

$$[a_K, b_K] = \left[\min_{K \subseteq S \subseteq V} L_S, \max_{K \subseteq S \subseteq V} L_S \right], \quad \text{osc}(K) = b_K - a_K. \quad (5)$$

The tables stay fixed while S varies. Completions can be disconnected, and need not fit the debate budget. Enlarging K shrinks the family of completions, so it cannot increase the oscillation.

Suppose a core contains 100 non-target vertices and every completion has log-odds between 1.2 and 1.3. A player with 100 moves can reveal the whole core, counting any vertices the opponent reveals toward the same goal. Max thereby guarantees at least 1.2; Min guarantees at most 1.3. The full-model answer belongs to the same interval.

Lemma 3.1 (Forcing a core). *If $|K| \leq h + 1$, then*

$$a_K \leq D_{2h} \leq b_K, \quad |D_{2h} - L_V| \leq \text{osc}(K). \quad (6)$$

Proof. On each move, Max reveals a remaining non-target vertex of K , if one exists, and otherwise makes any legal move. Every resulting terminal set contains K , so the strategy guarantees a_K . The identical construction for Min guarantees b_K . Finally $K \subseteq V$ implies $L_V \in [a_K, b_K]$. $\square$

The two forcing strategies are considered separately. Neither needs to be played at equilibrium. Connectivity was not used in the proof; it will let us organize influence by paths. The bound can be minimized over cores chosen after seeing the tables.

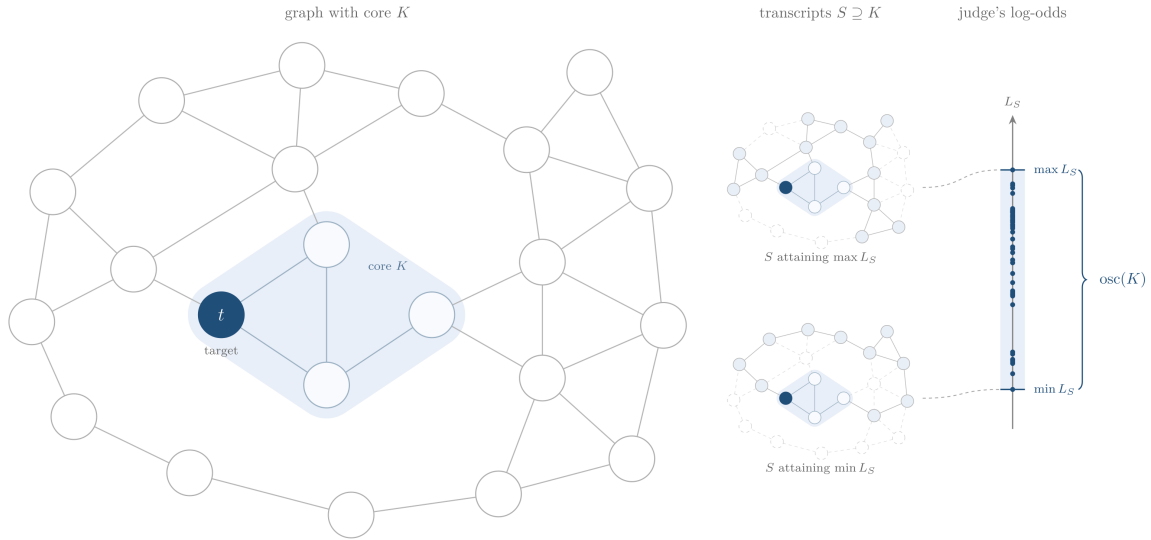


Figure 2: Several transcripts containing a common core. Oscillation is the full range over *all* such completions, including the full graph. A few pictured completions illustrate the definition without defining its extrema.

3.2 What the first experiments measure

For one board, let $O_h = \min_{|K| \leq h+1} \text{osc}(K)$, with the minimum over connected cores containing t . Then $|D_{2h} - L_V| \leq O_h$ board by board. The curve O_h is nonincreasing; the error of an alternating game need not be. A first successful budget is therefore not a theorem of convergence at every later budget.

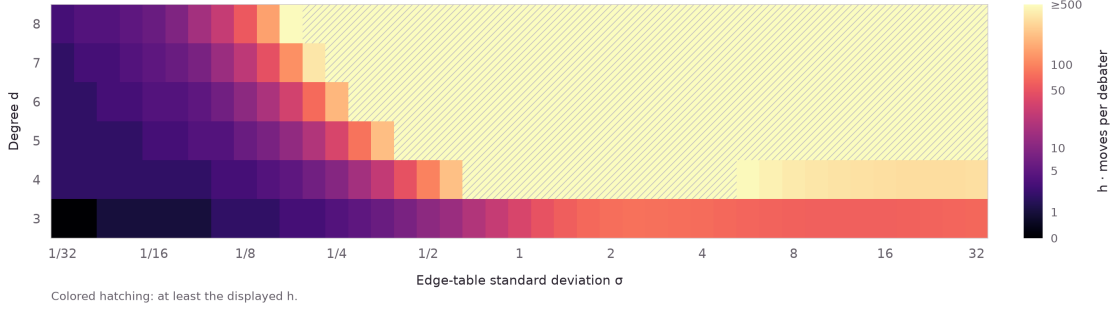


Figure 3: A numerical crossing of the *sample mean* of O_h at 0.1. Each cell uses 1,000 iid Gaussian-table draws on a 50,000-vertex partially filled regular tree, with zero unary factors. Colors use the upper numerical crossing, capped at 500. This is not the mean of the individual boards' crossing budgets, and is not a population confidence guarantee.

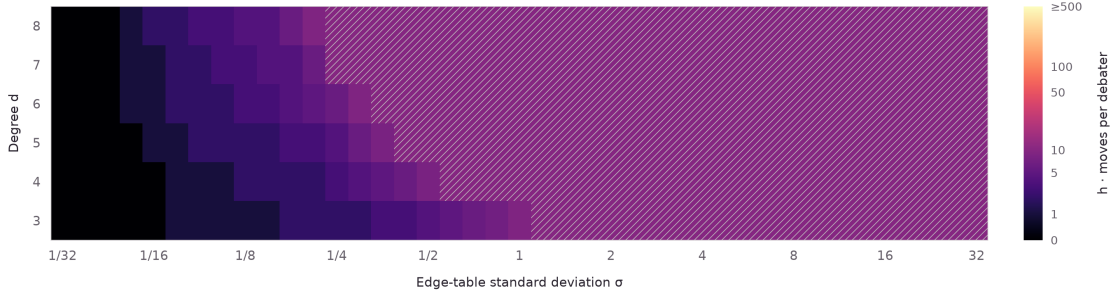


Figure 4: Integer debate budgets for target mean absolute error 0.1 on 50,000-vertex partially filled regular trees, matching the core panel. The 240 resolved cells retain simultaneous 95% confidence decisions, conditional on valid guarded numerical enclosures. Six boundary cells are assigned approximately: choose the smaller budget when its sample mean-error upper bound is at most 0.101, otherwise the larger budget. Hatching marks the cutoff $h \geq 10$ under this convention. The strict statistical brackets are retained separately in the figure data.

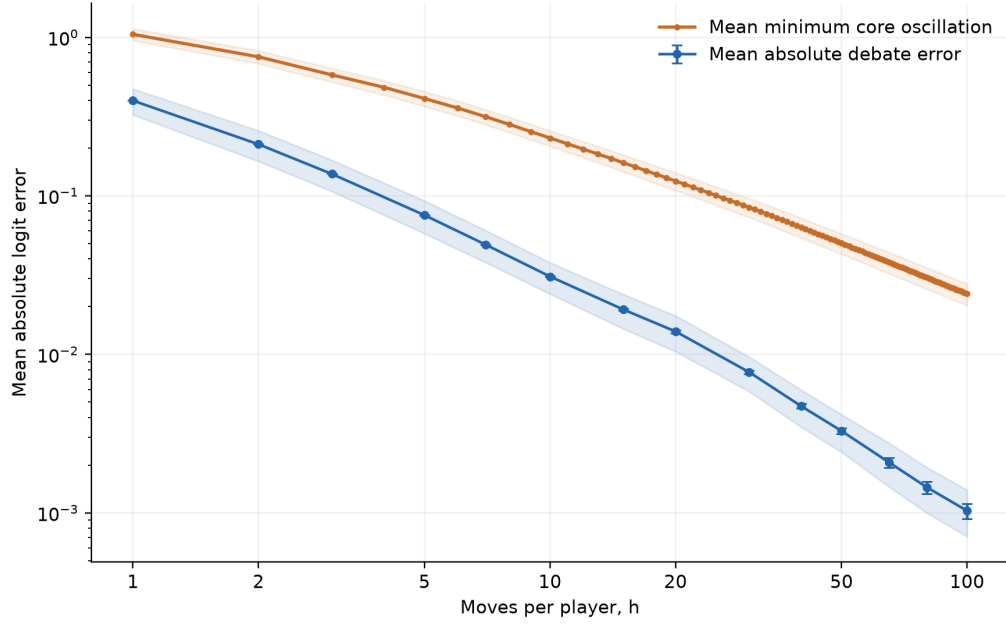


Figure 5: A paired comparison: degree 4, $\sigma = 0.35$, zero unary factors, and tree balls of depth 8 (13,121 vertices). The 64 draws use seeds 7–70. Blue is mean absolute debate error and orange is mean optimal-core oscillation on those same boards. Bars represent numerical enclosures; shading adds approximate pointwise 95% sampling uncertainty. A roughly straight segment on logarithmic axes suggests a power over the displayed range, without identifying an asymptotic exponent.

4 Factor-specific debate bounds

4.1 What one edge can transmit

To understand a core's boundary, first consider a single edge $e = \{u, v\}$. Every binary table has the expansion

$$\theta_{(u,v)}(x_u, x_v) = a_e + \frac{1}{2}H_{e,u}x_u + \frac{1}{2}H_{e,v}x_v + \frac{1}{2}J_ex_ux_v. \quad (7)$$

For example,

$$\begin{aligned} H_{e,u} &= \frac{1}{2}(\theta_{++} + \theta_{+-} - \theta_{-+} - \theta_{--}), \\ H_{e,v} &= \frac{1}{2}(\theta_{++} - \theta_{+-} + \theta_{-+} - \theta_{--}), \\ J_e &= \frac{1}{2}(\theta_{++} - \theta_{+-} - \theta_{-+} + \theta_{--}). \end{aligned}$$

The constant a_e cancels from odds. The two H 's are local fields contributed by the edge; J_e is its interaction. Our convention puts $J_e/2$ in the energy, which accounts for the factors of two below.

On a tree, remove e and let z be the log-odds at v on its side of the cut. Summing over x_v shows that the edge contributes to u the message

$$H_{e,u} + r_{J_e}(z + H_{e,v}), \quad r_j(x) = \log \cosh \frac{x+j}{2} - \log \cosh \frac{x-j}{2}. \quad (8)$$

For $j \geq 0$, differentiation gives

$$r'_j(x) = f_j(x) = \frac{\sinh j}{\cosh x + \cosh j}, \quad 0 \leq f_j(x) \leq f_j(0) = \tanh(j/2). \quad (9)$$

The response is odd in both x and j , so for either sign of j its Lipschitz constant is $c_e = \tanh(|J_e|/2)$. It takes values between $-|J_e|$ and $|J_e|$. Large $|x|$ makes its slope small: an already strongly biased input has little remaining capacity to change the output.

An absent edge contributes zero. A present edge with its far endpoint pinned contributes $H_{e,u} \pm J_e$. Therefore every possibility lies in

$$\mathcal{I}_{(u,v)} = [\min(0, H_{e,u} - J_e, H_{e,u} + J_e), \max(0, H_{e,u} - J_e, H_{e,u} + J_e)].$$

Its width is

$$\begin{aligned} I_{(u,v)} &= |J_e| + \max(|H_{e,u}|, |J_e|) \\ &= \frac{1}{2}|\theta_{(u,v)}(+, +) - \theta_{(u,v)}(-, +)| + \frac{1}{2}|\theta_{(u,v)}(+, -) - \theta_{(u,v)}(-, -)| + |J_e|. \end{aligned} \quad (10)$$

This follows from the identity that the range of three numbers equals half the sum of their three pairwise distances. I is directed because it measures the effect at a specified endpoint; J and c are not.

4.2 Routes, gates, and cycles

A route $P = (t, w_1, \dots, w_r)$ is a self-avoiding path: no vertex appears twice. Its length is r . The empty route consists of t alone. A *core routing* $\mathcal{R}$ is a set of routes contained in K , containing the empty route and every prefix of each of its members. The routing need not reach every vertex of K .

A *gate* is a pair $(P, (v, u))$ where P ends at v , u is not on P , and appending u gives a route outside $\mathcal{R}$. A gate may lead to a vertex still inside K if that extension was omitted from the routing.

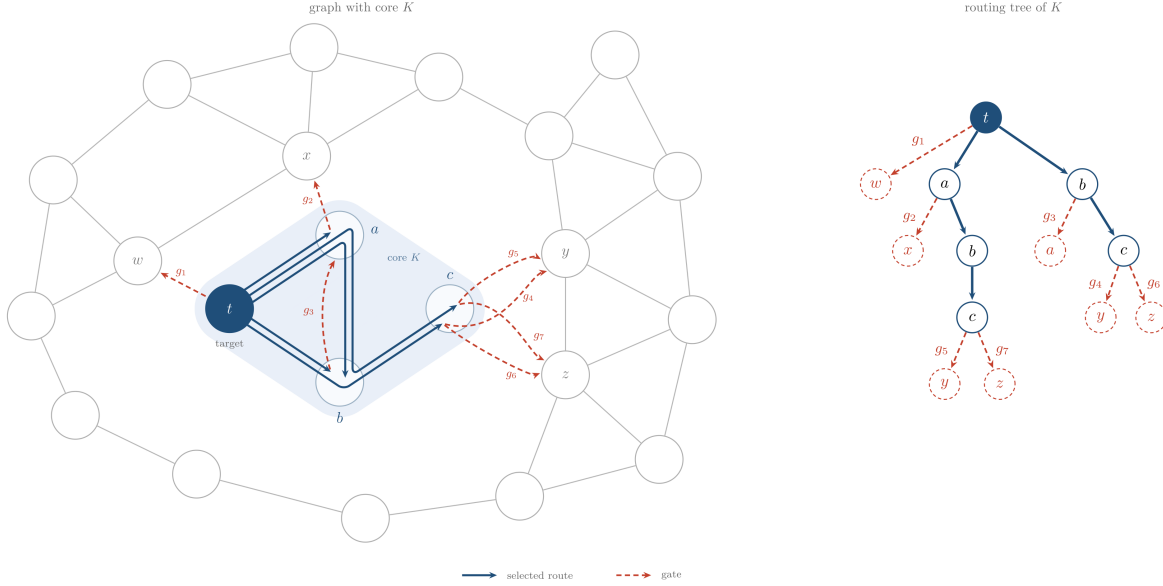


Figure 6: A routing viewed on the graph and as a tree of path histories. Different histories ending at the same physical vertex are different tree states. Red dashed arrows are omitted self-avoiding extensions. In general these include omitted internal extensions as well as exits from the core.

A physical edge may occur at several gates, after different histories. Cycle-closing steps are not gates.

Here is why a tree of histories can represent a graph with cycles exactly. Order the edges once and for all. To compute odds at v , first fix $x_v = -1$, then switch its incident factors one at a time to their $x_v = +1$ versions. The ratio of partition functions telescopes into one ratio per edge. In the ratio for $\{v, u\}$, delete v ; its other incident edges become unary factors at their other endpoints, with v pinned according to whether that edge has already been switched. The remaining ratio is exactly the one-edge calculation (8), using the cavity odds at u in that smaller model.

Repeat. Each ordinary recursive step deletes the previous vertex, so the ordinary states are self-avoiding paths. A return to an earlier vertex is replaced by its fixed pinned contribution. The recursion terminates because the graph is finite. This is the binary self-avoiding-walk construction associated with Weitz and its graphical-model extensions [3, 4]. The telescoping argument also shows why the same edge order gives compatible pins in every induced submodel containing a specified route. No independence of different history branches is asserted.

4.3 From boundary intervals to an error certificate

Truncate this recursion at the routing. Keep its internal messages and its fixed cycle pins; at each gate allow an independent parameter throughout $\mathcal{I}_{(v,u)}$. Every completion $S \supseteq K$ supplies one permissible choice of gate parameters: its true message if the extension is present, zero if absent. Thus the root interval of the relaxed recursion contains every L_S .

At an internal state, input widths add and the response multiplies their sum by at most c_e on the arrival edge. At the root, widths simply add. Propagating from the leaves toward the root therefore proves the following statement at the moment we have all its ingredients.

Theorem 4.1 (Core routing certificate). *For arbitrary finite tables and unary factors, every core K and routing $\mathcal{R}$ satisfy*

$$\text{osc}(K) \leq \sum_{(P,e) \in \text{gate}(\mathcal{R})} I_e \prod_{e' \in P} \tanh(|J_{e'}|/2). \quad (11)$$

If $|K| \leq h + 1$, the same expression bounds $|D_{2h} - L_V|$.

Proof. Let Δ_P be a nonroot state's output width. Its input is the sum of fixed terms, child messages, and gate parameters. Hence

$$\Delta_P \leq c_{\text{arrival}(P)} \left(\sum_{Q \text{ child of } P} \Delta_Q + \sum_{(P,e) \text{ gate}} I_e \right).$$

The corresponding root inequality has no leading c . Expanding it recursively gives exactly one summand per gate in (11). The completion containment just proved bounds $\text{osc}(K)$ by this root width; Lemma 3.1 then bounds debate error. $\square$

The gate edge contributes its reach, and is *not* also included in the product. Both the core and routing may depend on the realized tables. Taking all self-avoiding routes inside K leaves only gates exiting K . On a tree this is one route to each core vertex. On a loopy graph a chosen spanning tree alone does not suffice unless its omitted extensions are also charged.

The interval construction is an upper bound, not a formula for exact oscillation. For example, a single edge with $J \neq 0$ but zero H 's and unary fields has $L_{\{t\}} = L_V = 0$. Its singleton core has oscillation zero, whereas the relaxed reach has width $2|J|$.

5 Ensemble bounds

5.1 Random fields help attenuate influence

Now fix the graph and sample all four entries of each edge table independently from $N(0, \sigma^2)$, independently across edges. The unary factors are arbitrary and fixed. They may also be random with any joint law independent of all edge tables, since we can condition on them first. The three contrasts in (7) are orthogonal. Consequently $H_{e,u}, H_{e,v}, J_e$ are independent $N(0, \sigma^2)$ variables.

An input to a message contains several fresh Gaussian fields. Those fields usually shift the input away from the location where the response has its largest slope. Averaging the interval recursion exploits this shift. Simply replacing each c_e in (11) by its expectation would give a different and weaker calculation.

Define

$$g_\sigma = \frac{(2 + \sqrt{2})\sigma}{\sqrt{\pi}}, \quad \alpha_m(\sigma) = \mathbb{E} \frac{\sinh(\sigma|Z_0|)}{\cosh(\sqrt{m}\sigma Z_1) + \cosh(\sigma|Z_0|)}, \quad (12)$$

where Z_0, Z_1 are independent standard Gaussians and $m \geq 1$. The first quantity is the mean reach $\mathbb{E}I_e$: the two single differences in (10) have variance $2\sigma^2$, and J has variance σ^2 . The second quantity is the mean slope when the input contains an independent Gaussian of variance $m\sigma^2$.

Lemma 5.1 (Averaging an interval). *Let $T \sim N(0, m\sigma^2)$, and let $j \geq 0$. For every real z ,*

$$\mathbb{E}f_j(T + z) \leq \mathbb{E}f_j(T) =: c_m(j).$$

If an interval $[A, B]$ is independent of T , then, conditionally on A, B, j ,

$$\mathbb{E}_T[r_j(T + B) - r_j(T + A)] \leq c_m(j)(B - A).$$

Also $\mathbb{E}_J c_m(|J|) = \alpha_m(\sigma)$ for $J \sim N(0, \sigma^2)$, and α_m decreases with m .

Proof. The function f_j is even and decreases with $|x|$. Expressing it as a superposition of indicators of centered intervals reduces the first claim to the fact that a centered Gaussian puts most mass in an interval of a given length when that interval is centered at zero. Integrate this bound between A and B to obtain the second claim. Increasing the variance reduces the mass in each centered interval, proving the last assertion. $\square$

We must check where the fresh Gaussian comes from, especially in a graph with cycles. Fix a routing independently of the tables. At a state P ending at v , let $m(P)$ be the number of incident edges that are *not* gate edges there. This includes the arrival edge, ordinary children, and cycle pins. Their fields at v sum to an independent $N(0, m(P)\sigma^2)$ variable. Once these fields are removed, the remainder of the input interval does not use them: a strict descendant cannot revisit v . It also cannot reuse the arrival coupling, whose two endpoints are already on the route. Gate fields are kept inside their gate intervals and are not counted as fresh.

Apply Lemma 5.1 to this input interval, then average the independent arrival coupling. If P has children Q and b_P gates,

$$\mathbb{E}\Delta_P \leq \alpha_{m(P)}(\sigma) \left(\sum_Q \mathbb{E}\Delta_Q + b_P g_\sigma \right). \quad (13)$$

Different children can reuse physical variables. Only their individual expectations are added in this equation; none are multiplied together. Unrolling gives

$$\mathbb{E} \text{osc}(K) \leq g_\sigma \sum_{(P,e) \in \text{gate}(\mathcal{R})} \prod_{\emptyset \neq P' \preceq P} \alpha_{m(P')}(\sigma). \quad (14)$$

Unlike the deterministic bound, this averaged formula assumes the routing was fixed without looking at the tables. Substituting a data-dependent optimal routing would invalidate the fresh-variable argument.

5.2 A radius becomes a budget

For $d \geq 3$, put $q = d - 1$ and define

$$\rho = \rho(d, \sigma) = \max_{2 \leq m \leq d} (m - 1)\alpha_m(\sigma). \quad (15)$$

An intermediate vertex of physical degree m has at most $m - 1$ self-avoiding continuations and average attenuation α_m . Their combined cost is therefore at most ρ . The maximum over m is what makes the argument uniform over graphs with nonconstant degree.

The ball $B_R(t)$ contains at most

$$1 + d \sum_{j=0}^{R-1} q^j = 1 + d \frac{q^R - 1}{d - 2}$$

vertices. Therefore the affordable radius is

$$R_h = \left\lfloor \frac{\log(1 + (d - 2)h/d)}{\log q} \right\rfloor. \quad (16)$$

Use the routing of all paths of length at most R_h . Its gates occur at depth R_h . The last state has at most q gates and at least one fresh field, so it costs at most $q\alpha_1 g_\sigma$. Every earlier nonroot generation multiplies the sum by at most ρ , while the root has at most d branches.

Theorem 5.2 (Gaussian mean-error bound). *Let G have maximum degree at most $d \geq 3$, and let $h \geq 1$ with $2h \leq |V| - 1$. If $\rho(d, \sigma) < 1$, set*

$$\gamma = \frac{\log(1/\rho)}{\log(d - 1)}, \quad C(d, \sigma) = \alpha_1(\sigma) g_\sigma \frac{d(d - 1)}{\rho^2} \left(\frac{d}{d - 2} \right)^\gamma.$$

Then, uniformly in graph size, target, and fixed unary factors,

$$\boxed{\mathbb{E}|D_{2h} - L_V| \leq C(d, \sigma) h^{-\gamma}.} \quad (17)$$

For $R_h \geq 1$ a sharper finite-budget version is

$$\mathbb{E}|D_{2h} - L_V| \leq d g_\sigma \min\{1, q\alpha_1 \rho^{R_h - 1}\};$$

for $R_h = 0$, use $d g_\sigma$.

Proof. The preceding path count and (14) give the second bound; the singleton core gives $d g_\sigma$. Since $R_h - 1 \geq \log_q(1 + (d - 2)h/d) - 2$, the path count for $R_h \geq 1$ gives

$$\rho^{R_h - 1} \leq \rho^{-2} (1 + (d - 2)h/d)^{-\gamma} \leq \rho^{-2} (d/(d - 2))^\gamma h^{-\gamma}.$$

When $R_h = 0$, $h < d$. To check the same constant, let m maximize ρ . Then $\rho = (m - 1)\alpha_m \leq q\alpha_1$, and

$$\frac{C h^{-\gamma}}{d g_\sigma} \geq \rho^{-1} (d/((d - 2)h))^\gamma = \left(\frac{q d}{(d - 2)h} \right)^\gamma > 1.$$

Thus the stated power also covers small budgets. $\square$

The hypothesis $\rho < 1$ is essential to the *decaying* power-law statement. When it fails, it is the certificate that fails; this argument has not established nonlocality or nonconvergence.

There is a useful all-scale conclusion at degrees three and four. The identity $r'_j = f_j$ also gives

$$f_j(x) = \frac{1}{2} \left(\tanh \frac{x+j}{2} - \tanh \frac{x-j}{2} \right).$$

If Λ has logistic distribution function $(1 + e^{-x})^{-1}$, this implies

$$\alpha_m(\sigma) = \mathbb{P}(|\sqrt{m}Z_1 - \Lambda/\sigma| \leq |Z_0|).$$

For any fixed value of $|Z_0|$, adding an independent shift to the centered Gaussian cannot increase its mass in that centered interval. Hence

$$\alpha_m(\sigma) < a_m := \mathbb{P}(\sqrt{m}|Z_1| \leq |Z_0|) = \frac{2}{\pi} \arctan \frac{1}{\sqrt{m}}, \quad (18)$$

with equality in the limit $\sigma \rightarrow \infty$. The last expression follows from the angular measure of the corresponding wedges in the plane. The function $(m-1)a_m$ increases with m ; differentiating $(x-1)\arctan(x^{-1/2})$ verifies this. Therefore

$$\rho(3, \sigma) < \frac{2}{3}, \quad \rho(4, \sigma) < \frac{6}{\pi} \arctan \frac{1}{2} < 0.886.$$

The mean-error theorem holds at every finite scale for these degrees. At any fixed degree it also holds for sufficiently small σ , since $\alpha_m(\sigma) \rightarrow 0$ as $\sigma \downarrow 0$.

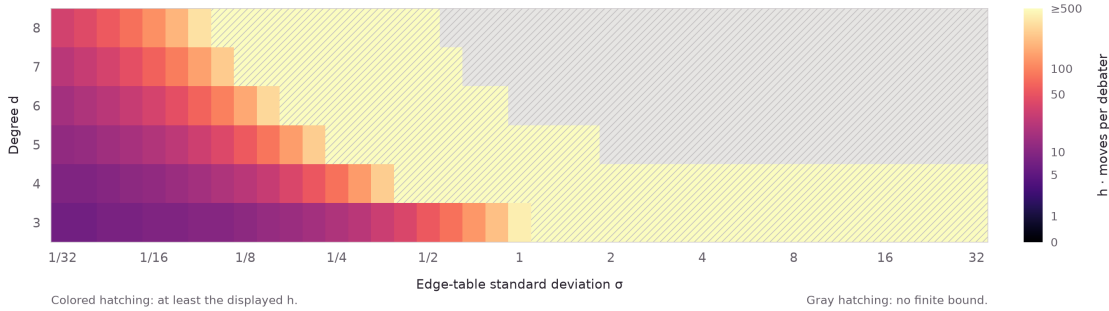


Figure 7: The smallest integer $h \geq 1$ for which (17) is at most 0.1, evaluated by numerical quadrature. Colored hatching means a sufficient budget at least 500; gray hatching means $\rho \geq 1$, so this formula supplies no decaying guarantee. The calculation is uniform over all graph sizes; it is not a simulation on a particular tree.

5.3 High noise: what a fractional moment says

The mean-error argument sees some saturation, but it does not exploit all of it. A stronger calculation at large σ controls

$$\|D_{2h} - L_V\|_p = (\mathbb{E}|D_{2h} - L_V|^p)^{1/p}, \quad 0 < p < 1.$$

For such p this is a quasi-norm. It gives less weight to rare large errors than the mean does. Appendix A proves a fully explicit bound $\|D_{2h} - L_V\|_p \leq U(d, \sigma, h, p)$, valid on *every* graph of

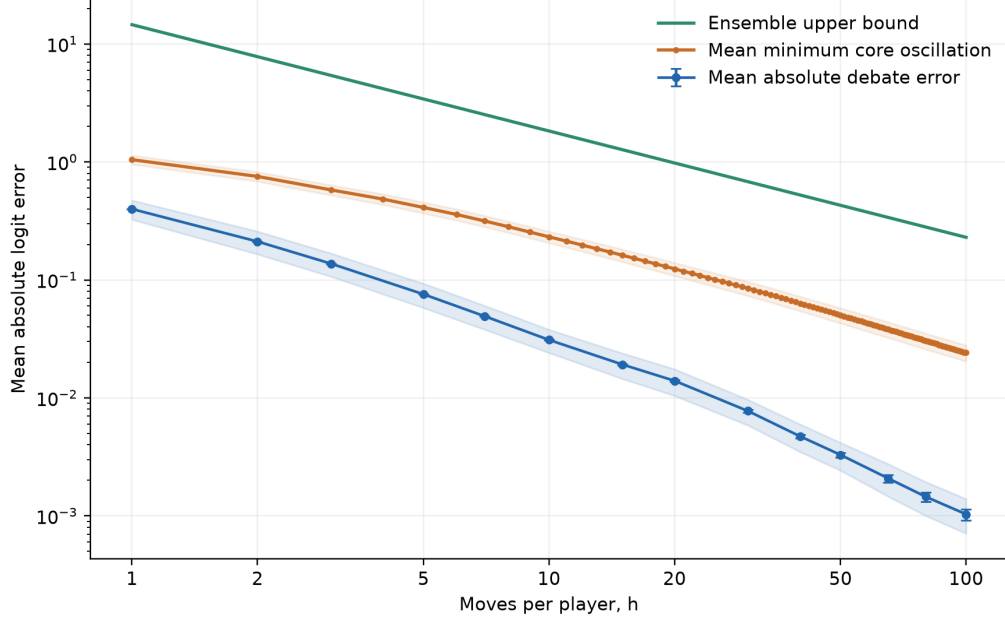


Figure 8: The paired experiment of Figure 5, with the uniform ensemble bound $14.57957h^{-0.9013297}$ added. Its constant and exponent are calculated from the theorem without fitting to the data. Similar-looking slopes over this budget range do not establish equal asymptotic exponents.

maximum degree three or four, including graphs with arbitrary cycles. It yields, for any specified failure probability $\eta \in (0, 1)$,

$$\mathbb{P}(|D_{2h} - L_V| > \eta^{-1/p} U(d, \sigma, h, p)) \leq \eta. \quad (19)$$

At degree three, for example, $\sigma > 6$ and $p = 6/\sigma$ give a fixed-confidence threshold

$$O_{\sigma, \eta}(h^{-\sigma \log(5/4)/(6 \log 2)}).$$

The exponent improves with σ , but the constants can be large. Changing p also changes the error metric; fixed confidence is the common criterion in this comparison.

This does not establish a corresponding improvement of the mean-error exponent. It does not establish a heavy-tailed error distribution either. In fact there is a Gaussian tail envelope. If $A_e = \theta_{(t,u)}(+, -) - \theta_{(t,u)}(-, -)$ and $B_e = \theta_{(t,u)}(+, +) - \theta_{(t,u)}(-, +)$, then

$$|D_{2h} - L_V| \leq W := \sum_{e \ni t} [\max(0, A_e, B_e) - \min(0, A_e, B_e)]. \quad (20)$$

Each summand is the maximum of $A_e, -A_e, B_e, -B_e, A_e - B_e, B_e - A_e$. Thus W is the maximum of at most 6^d centered Gaussian linear forms, each of variance at most $4d\sigma^2$. A Gaussian Chernoff bound and a union bound give

$$\mathbb{P}(|D_{2h} - L_V| > u) \leq \min\{1, 6^d e^{-u^2/(8d\sigma^2)}\}, \quad u > 0. \quad (21)$$

All moments exist at fixed d, σ . Rare errors can still matter for estimating a mean; the tail's mathematical class is a separate issue.

6 Lower bounds

6.1 Why an inspection budget leaves information behind

An upper bound supplies a strategy. A lower bound must allow the players to choose their strategy after seeing the entire board. Treating omitted vertices as an independent random sample would miss that difficulty.

Work now on a regular degree- d tree ball of depth L , with zero unary factors. The root has d children; every other nonleaf has $q = d - 1$ children. All edge-table entries are independent $N(0, \sigma^2)$. At depth $n + 1$ there are dq^n edges. We will build an orthonormal family of random test functions, one per edge on that level. The full-model answer correlates with all of them. A budgeted judge has a bound on the *sum* of its correlations, even though its selected transcript depends on all the tables. The deficit gives an RMS error by orthogonal projection.

The distinction between a fixed tree and a family of trees matters. A fixed finite model can eventually be fully revealed. A polynomial lower bound as h grows will therefore require a depth growing with $\log h$. We will state that depth explicitly.

6.2 Tests adapted to a path

Orient messages toward the root, writing the edge message as

$$m_e = M_e + r_{J_e}(X_e), \quad X_e = U_e + \sum_{f \text{ child of } e} m_f.$$

M_e is the field at the parent endpoint, U_e the field at the child; all M_e, U_e, J_e are independent $N(0, \sigma^2)$ variables. Let $c_e = \tanh(|J_e|/2)$. For an edge e at depth $n + 1$, write $f \prec e$ for its n strict ancestor edges.

To include alternating game values without making false claims about arbitrary selection rules, let $\mathcal{V}_k$ be the class generated by terminal logits whose target components contain at most k non-target vertices, using finite pointwise minima, maxima, and *fixed* convex combinations. All finite perfect-information reveal games with total budget k lie in this class, as do fixed averages of their values. Arbitrary discontinuous board-dependent selections, or convex weights depending on the board, are not automatically included.

Choose a nonnegative bounded function $w(J)$ with $\mathbb{E}w(J)^2 = 1$, and define

$$\Psi_e = \frac{M_e}{\sigma} \prod_{f \prec e} \text{sgn}(J_f) w(J_f), \quad A_n = \mathbb{E} \prod_{f \prec e} |m'_f(X_f)| w(J_f), \quad C_e = \prod_{f \prec e} c_f w(J_f).$$

Here $m'_f(X_f)$ means $r'_{J_f}(X_f)$. The number A_n is the same for every edge on the level by symmetry. If $c_f w(J_f) \leq K$ always, then $C_e \leq K^n$.

Lemma 6.1 (A budgeted projection). *For every $V \in \mathcal{V}_k$,*

$$\|V - L_V\|_2 \geq \frac{\sigma}{\sqrt{dq^n}} (dq^n A_n - kK^n)_+. \quad (22)$$

Here and below $x_+ = \max(x, 0)$.

Proof. The Ψ_e have norm one. For two different edges on the same level, one factor M_e is centered and independent of every other factor in the product, so their inner product is zero. Hence they are

orthonormal. Gaussian integration by parts in M_e gives

$$\langle L_V, \Psi_e \rangle = \sigma \mathbb{E} \left[\frac{\partial L_V}{\partial M_e} \prod_{f \prec e} \text{sgn}(J_f) w(J_f) \right] = \sigma A_n.$$

The chain rule supplied the product of actual slopes; the sign factors make it nonnegative.

For a fixed terminal transcript the derivative is zero unless the edge belongs to its target component, and otherwise is a product of the slopes along its path. At most k level edges can contribute. After multiplication by the sign and weight factors, every contribution is between zero and K^n . Finite minima, maxima, and fixed convex combinations are Lipschitz in the Gaussian fields. Almost everywhere their gradients are convex combinations of terminal gradients. The same pointwise bound on the sum therefore survives, and integration by parts gives

$$0 \leq \sum_e \langle \Psi, \Psi_e \rangle \leq \sigma k K^n.$$

The sum of the full-model coefficients is $\sigma d q^n A_n$. Bessel's inequality followed by Cauchy-Schwarz over the $d q^n$ coefficients proves (22). $\square$

6.3 A scalar inequality that survives the whole path

Normalize fields by σ , and set

$$\begin{aligned} c(z) &= \tanh(\sigma|z|/2), \\ r(z, x) &= \frac{2}{\sigma} \operatorname{artanh} \left(\tanh \frac{\sigma z}{2} \tanh \frac{\sigma x}{2} \right), \\ \psi(z, x) &= \frac{1 + \cosh(\sigma z)}{\cosh(\sigma x) + \cosh(\sigma z)}. \end{aligned} \tag{23}$$

σr is the response, and ψ is its actual slope divided by the largest possible slope $c(z)$. The response is symmetric in law, because changing the sign of J changes its sign. Also $|r(z, x)| \leq |z|$, so its second moment is at most one.

Let $f : \mathbb{R} \rightarrow [0, 1]$ be even, with $0 < N := \mathbb{E}[f(Z)^2/c(Z)^2] < \infty$. Our choices below make the quotient bounded. Put

$$K = N^{-1/2}, \quad w(J) = K \frac{f(J/\sigma)}{c(J/\sigma)}.$$

Then $\mathbb{E} w^2 = 1$ and $cw \leq K$. The actual weighted slope on a path is $K f(Z) \psi(Z, X)$.

We need to average products of these slopes even though successive cavity inputs are dependent. The following inequality does that work. Put $\tau = d + 2/\sigma^2$. For every z, x ,

$$\psi(z, x) e^{-r(z, x)^2/(2\tau)} \geq e^{-\sigma^2 x^2/4}. \tag{24}$$

To verify it, let $C = \cosh(\sigma z)$, $t = \cosh(\sigma x)$, and $D = (1 + Ct)/(t + C)^2$. Direct differentiation gives

$$(\log \psi)_{xx} = -\sigma^2 D, \quad r_x^2 = \frac{C^2 - 1}{(t + C)^2}, \quad r r_{xx} \leq 0, \quad 2D + r_x^2 \leq 1.$$

Since $1/\tau \leq \sigma^2/2$, the logarithm on the left of (24) has second derivative at least $-\sigma^2/2$. It is even and equals zero at the origin. Integrating this derivative bound twice proves the inequality.

Let Y be the response on the continuing child edge. Conditional on its subtree and its coupling, the normalized input one step nearer the root is

$$\sqrt{d}G + S + Y,$$

where G is standard Gaussian and S is the sum of the other $q - 1$ child responses. These quantities and the current coupling Z are independent; S is symmetric and $\mathbb{E}S^2 \leq q - 1$. Gaussian integration in (24) gives

$$\mathbb{E}_{Z,G}[f(Z)\psi(Z, \sqrt{d}G + x)e^{-r(Z, \sqrt{d}G + x)^2/(2\tau)}] \geq \frac{\mathbb{E}f(Z)}{\sqrt{1 + d\sigma^2/2}}e^{-x^2/(2\tau)}.$$

Furthermore, symmetry and Jensen's inequality imply

$$\begin{aligned}\mathbb{E}e^{-(S+y)^2/(2\tau)} &= e^{-y^2/(2\tau)}\mathbb{E}[e^{-S^2/(2\tau)}\cosh(Sy/\tau)] \\ &\geq e^{-y^2/(2\tau)}e^{-(q-1)/(2\tau)}.\end{aligned}$$

Thus if

$$\lambda = \frac{\mathbb{E}f(Z)}{\sqrt{1 + d\sigma^2/2}}e^{-(q-1)/(2\tau)}, \quad v = e^{-1/(2\tau)}, \quad (25)$$

conditional averaging of one step sends the auxiliary function $e^{-y^2/(2\tau)}$ to at least λ times itself. Iterate from the root down the path. At its last response, Jensen gives an expectation at least v . Since the auxiliary function is at most one, we have proved

$$A_n \geq v(K\lambda)^n. \quad (26)$$

The auxiliary Gaussian is what keeps the dependence along the path under control. Factoring the actual gains into independent one-edge expectations would not be justified.

6.4 An explicit lower bound

All the ingredients are now in place. The next statement is deliberately shorter to calculate than the stronger certificates in Appendix B.

Theorem 6.2 (An explicit RMS lower bound). *Let $d \geq 3$, $\sigma > 0$, and $q = d - 1$. Define*

$$A_0 = \frac{q \exp(-(d-2)\sigma^2/(4 + 2d\sigma^2))}{\sqrt{1 + d\sigma^2/2}}.$$

Suppose $A_0 > 1$. Set

$$\begin{aligned}\varepsilon &= \tanh\left(\frac{\sigma}{4}(1 - A_0^{-1})\right), & C(Z) &= \tanh(\sigma|Z|/2), & F(Z) &= \min\{1, C(Z)^2/\varepsilon^2\}, \\ a &= A_0\mathbb{E}F(Z), & N &= \mathbb{E}\frac{F(Z)^2}{C(Z)^2}, & \delta &= \frac{\log(qN)}{2\log a} - 1,\end{aligned}$$

with Z standard Gaussian and the ratio defining N continued by zero at $Z = 0$. Then $a > 1$ and $\delta > 0$. For every integer $h \geq 1$, on the Gaussian regular tree ball described above with

$$L \geq 1 + \left\lceil \frac{\log(2h)}{\log a} \right\rceil, \quad (27)$$

and legal budget $2h \leq |V| - 1$, ordinary alternating debate satisfies

$$\boxed{(\mathbb{E}|D_{2h} - L_V|^2)^{1/2} \geq \sigma \left(\sqrt{d} e^{-1/(2d)} - \frac{1}{\sqrt{d}} \right) (2ah)^{-\delta}.} \quad (28)$$

The same bound holds for every value in $\mathcal{V}_{2h}$.

Proof. Put $u = (1 - A_0^{-1})/2$. The event $C(Z) < \varepsilon$ is $|Z| < u$. The density of $|Z|$ is less than one, so $1 - \mathbb{E}F \leq u$ and $a \geq (A_0 + 1)/2 > 1$. The quotient in N is bounded by ε^{-2} . Cauchy–Schwarz and $\mathbb{E}C^2 \leq \sigma^2/4$ imply

$$\frac{a^2}{qN} \leq \frac{A_0^2}{q} \mathbb{E}C^2 \leq \frac{q\sigma^2}{4 + 2d\sigma^2} < \frac{1}{2}.$$

Consequently $\delta > 0$ and $a/\sqrt{qN} = a^{-\delta}$.

Use $f = F$ in (25). Then $q\lambda = a$. Substituting (26) in Lemma 6.1 gives, for every available level $n + 1$,

$$\|D_{2h} - L_V\|_2 \geq \sigma \sqrt{d} v a^{-\delta n} \left(1 - \frac{2h}{dva^n} \right)_+. \quad (29)$$

Choose $n = \lceil \log_a(2h) \rceil$, which is allowed by (27). Then $a^n \geq 2h$, and $a^{-\delta n} \geq (2ah)^{-\delta}$. Since $\tau \geq d$, $\sqrt{d}v - 1/\sqrt{d} \geq \sqrt{d}e^{-1/(2d)} - 1/\sqrt{d} > 0$. This proves the result. $\square$

The square on ε in F matters: it makes the threshold $C < \varepsilon$, which is exactly the small-ball estimate used in the proof. The theorem gives a lower bound on RMS, while the preceding ensemble theorem bounded mean absolute error. Those are different metrics. An RMS lower bound alone does not imply an equally large mean lower bound. Appendix C performs a separate tail conversion.

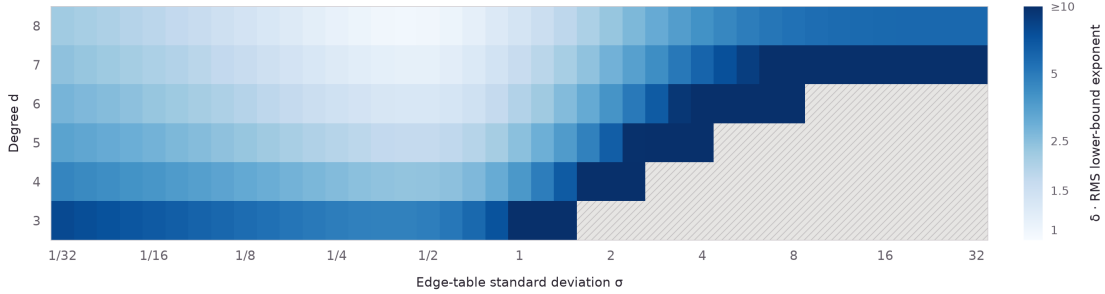


Figure 9: RMS lower-bound exponents from the stronger certificate family in Appendix B. The 193 populated cells are numerical quadrature evaluations, not outward-certified decimal inequalities or optimal exponents. Smaller δ means slower decay and a stronger asymptotic obstruction; prefactors and required depth still matter. Hatched cells fail the condition $A_0 > 1$ of Theorem 6.2 and were not evaluated with the stronger certificates; this does not show that polynomial lower bounds are impossible there.

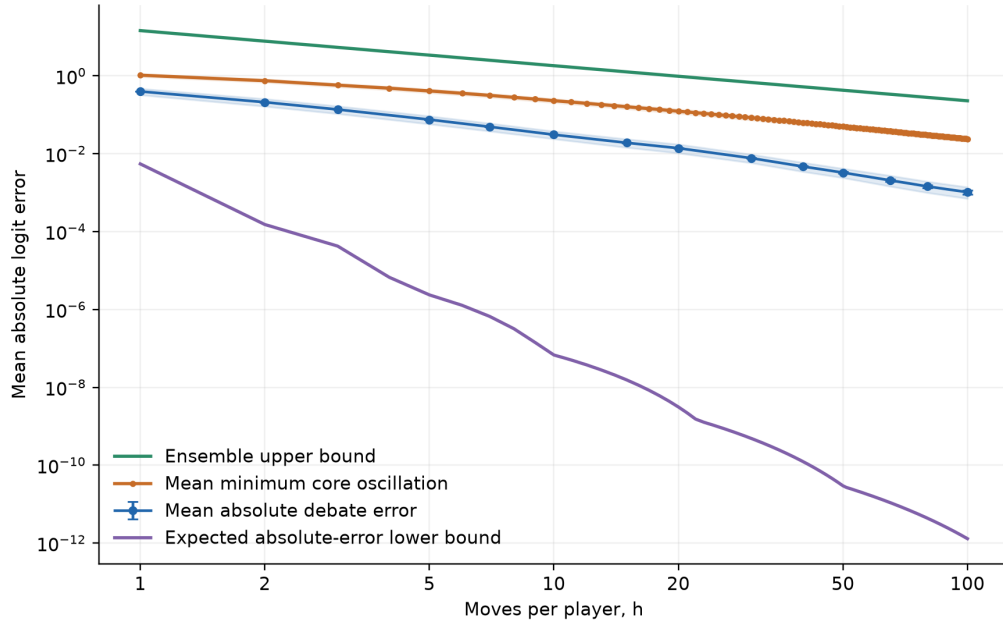


Figure 10: The same finite-depth experiment as Figure 5, now with a population *mean absolute-error* lower bound in purple. It is obtained by optimizing (29) over the eight available levels using stronger constants, then applying the tail conversion (38). At $h = 1, 10, 100$ its values are approximately 0.00550, $6.79 \cdot 10^{-8}$, and $1.30 \cdot 10^{-12}$. It is neither an RMS curve nor a confidence limit for the empirical mean. The quadrature is numerical, without outward rounding.

7 Scope of the conclusions

The forcing argument is insensitive to the order of the players' moves; what it uses is that either player can fit a useful core into its own quota. The remaining difficulty is estimating how stable a small core can be. On the boards of Figure 5, the mean optimized-core oscillation at $h = 100$ is about 0.024, while the uniform bound of Theorem 5.2 evaluates to about 0.23 there: a core chosen after seeing the tables can be much better than the radius bound predicts.

The lower bound concerns the induced-submodel judge and the class $\mathcal{V}_k$ of Section 6; it is not an impossibility theorem for other judges or protocols. Likewise $\rho < 1$ is a sufficient condition. Neither a gray heatmap cell nor a run stopped at a budget cutoff shows that debate fails there.

A An explicit high-noise fractional-moment bound

This appendix supplies the quantitative statement used in Section 5.3. Its proof uses marginal moments of interval widths, so cycles introduce no independence assumption between siblings.

Theorem A.1 (Fractional moment and confidence). *Under the assumptions of Theorem 5.2, let $d \in \{3, 4\}$ and $h \geq d$. Choose p in the indicated interval:*

d	$\bar{\rho}_d$	b_d	allowed p
3	2/3	4/5	$12/(5\sigma) < p < 1$
4	89/100	27/25	$108/(11\sigma) < p < 1$

Put $n = R_h - 1 \geq 0$, with R_h as in (16), and

$$\lambda = \bar{\rho}_d + \frac{b_d}{\sigma p}, \quad \mu = \bar{\rho}_d^{1-p/3}, \quad S_n = \sum_{j=0}^{n-1} \lambda^{n-1-j} \mu^j.$$

The sum is zero for $n = 0$. For $n \geq 1$ it equals $(\lambda^n - \mu^n)/(\lambda - \mu)$ when $\lambda \neq \mu$, and $n\lambda^{n-1}$ when $\lambda = \mu$. Then

$$\|D_{2h} - L_V\|_p \leq U(d, \sigma, h, p) := \sigma \left\{ d \left[\frac{8}{5} \lambda^n + \frac{3}{2} (d-1)^2 S_n \right] \right\}^{1/p}. \quad (30)$$

The confidence statement (19) follows for every $\eta \in (0, 1)$.

Proof. Normalize fields and logits by σ . A coupling $j \sim N(0, 1)$ has normalized response $r(j, x)$ as in (23), with absolute slope

$$F_{\sigma, j}(x) = \frac{\sinh(\sigma|j|)}{\cosh(\sigma x) + \cosh(\sigma|j|)} \leq e^{-\sigma(|x| - |j|)_+}.$$

Truncate the walk recursion at depth R_h . At each terminal state allow its cavity input to range over all real numbers. Its response interval has width $W_0 = 2|j|$. At an interior state, let W_r denote the output width after r layers have been propagated. Always $W_r \leq 2|j|$.

Suppose the state has k ordinary children and physical degree m . Then $k \leq m - 1 \leq q$. Conditional on all child intervals and cycle-pin shifts, their input is a fixed interval I of length $u = \sum_{i=1}^k W_{r,i}$, translated by a fresh $T \sim N(0, m)$. The arrival coupling j is also fresh. This is precisely the local independence established before (13). When $k = 0$ the width is zero. Otherwise $m \geq 2$ and

$$W_{r+1} \leq u \sup_{x \in T+I} F_{\sigma, j}(x).$$

By Lemma 5.1 and (18), $\mathbb{E}[W_{r+1} \mid I] \leq a_m u$.

For an even decreasing function $g : \mathbb{R} \rightarrow [0, 1]$, allowing its argument to range over an interval of length u enlarges each superlevel interval by u . The Gaussian density is at most $1/\sqrt{2\pi m}$, and translations of a centered interval only reduce its mass. Layer-cake integration gives

$$\mathbb{E}_T \sup_{x \in T+I} g(x) \leq \mathbb{E}_T g(T) + \frac{u}{\sqrt{2\pi m}}.$$

Use $g = F_{\sigma, j}^p$ and average j . If $A_m(c) = \mathbb{E}e^{-c(\sqrt{m}|Z_1| - |Z_0|)_+}$, this yields

$$\mathbb{E}[W_{r+1}^p \mid I] \leq u^p \left(A_m(\sigma p) + \frac{u}{\sqrt{2\pi m}} \right). \quad (31)$$

The positive part of $\sqrt{m}|Z_1| - |Z_0|$ has density on $x > 0$

$$g_m(x) = \sqrt{\frac{8}{\pi(m+1)}} e^{-x^2/(2(m+1))} \Phi_N \left(-\frac{x}{\sqrt{m(m+1)}} \right),$$

where Φ_N is the standard Gaussian distribution function. Both factors decrease for $x > 0$. Its density is therefore at most $g_m(0) = \sqrt{2/(\pi(m+1))}$, so

$$A_m(c) \leq a_m + \frac{1}{c} \sqrt{\frac{2}{\pi(m+1)}}.$$

The tabulated constants satisfy, for $1 \leq k \leq q$,

$$ka_{k+1} \leq \bar{\rho}_d, \quad k \sqrt{\frac{2}{\pi(k+2)}} \leq b_d.$$

Both left sides increase with k . For $d = 3$ use $a_3 = 1/3$ and $\sqrt{2/\pi} < 4/5$; for $d = 4$ use $3a_4 < 89/100$ and $3\sqrt{2/(5\pi)} < 27/25$. These inequalities follow, for example, from $\pi > 157/50$ and the alternating series for $\arctan(1/2)$. Thus $kA_m(\sigma p) \leq \lambda < 1$.

Let $w = \mathbb{E}W_0 = \sqrt{8/\pi} < 8/5$. Induction on mean widths and the pointwise bound $W_r \leq 2|j|$ give

$$\mathbb{E}W_r \leq w\bar{\rho}_d^r, \quad \mathbb{E}W_r^4 \leq 48.$$

Hölder interpolation, with $1 + p = (1 - p/3) \cdot 1 + (p/3) \cdot 4$, gives

$$\mathbb{E}W_r^{1+p} \leq (\mathbb{E}W_r)^{1-p/3} (\mathbb{E}W_r^4)^{p/3} \leq 5\mu^r.$$

Let y_r bound all p -moments at level r uniformly over states and graphs. Since $u^p \leq \sum_i W_{r,i}^p$ and $u^{1+p} \leq k^p \sum_i W_{r,i}^{1+p}$, (31) implies

$$y_{r+1} \leq \lambda y_r + \frac{3}{2} q^2 \mu^r, \quad y_0 \leq w^p \leq 8/5.$$

We used $1/\sqrt{2\pi m} < 3/10$ and $k^{1+p} \leq q^2$. Solving this scalar recurrence gives $y_n \leq (8/5)\lambda^n + (3/2)q^2 S_n$.

Every state at depth less than R_h has all incident edges in $B_{R_h}(t)$. Its fields and pins are fixed across completions of that core. All remaining variation enters through the terminal response intervals. The normalized root width is at most a sum of d widths of type W_n . Subadditivity therefore bounds the p -moment of debate error by $dy_n \sigma^p$. The core fits one player's budget by (16), which proves (30). Markov's inequality proves the confidence statement. $\square$

For $d = 3$, $\sigma > 6$, and $p = 6/\sigma$, we have $\lambda = 4/5$ and $\mu < 4/5$. Then $S_n = O_\sigma((4/5)^n)$ and $n = \log_2 h + O(1)$, giving the exponent in Section 5.3. At $d = 4$, $\sigma > 20$ and $p = 20/\sigma$ give $\lambda = 118/125$, $\mu < 118/125$, and fixed-confidence exponent $\sigma \log(125/118)/(20 \log 3)$. These are sufficient guarantees, not optimal thresholds for the allowed powers or scales.

B Stronger lower-bound certificates used by the figures

The level projection in Section 6 is reusable. To improve its constants, we improve the variance bound on a child response and the one-step Gaussian inequality. This appendix specifies the complete family used by the retained numerical figures, without treating numerical quadrature as a proof of rounded decimal values.

B.1 Bounding the response variance

Let $q = d - 1$ and define

$$F_0(v) = \mathbb{E} \log \cosh(\sqrt{v}Z/2), \quad D_d = \frac{2}{\pi}[\sqrt{d} + (1-d) \arctan(1/\sqrt{d})].$$

Let $b_f \in (0, 1)$ and $b_u \in (1/2, 1)$ be the solutions of

$$b_f = 1 - e^{-2F_0(\sigma^2(d+qb_f))}, \quad b_u = 1 - D_d e^{-qb_u/(2d)}, \quad b = \min(b_f, b_u). \quad (32)$$

Every normalized subtree response Y satisfies $\mathbb{E}Y^2 \leq b$. Here is a proof, including the fixed-point justification.

For the first bound, F_0 is increasing and concave: the derivative of $\log \cosh(\sqrt{v}/2)$ decreases with v . Conditional on its input X , a response $r_J(X)$ is symmetric and its modulus is at most $|J| \tanh(|X|/2)$. For any convex test function it is therefore dominated by the centered Gaussian $J \tanh(|X|/2)$. Replace independent child responses one at a time in the convex function $\log \cosh$. If each child satisfies $\mathbb{E} \tanh^2(X_i/2) \leq \beta$, concavity of F_0 gives

$$\mathbb{E} \log \cosh(X/2) \leq F_0(\sigma^2(d + q\beta)), \quad \mathbb{E} \tanh^2(X/2) \leq 1 - e^{-2F_0(\sigma^2(d+q\beta))}.$$

The last step is Jensen applied to e^{-2x} . The right-hand map is increasing and concave, positive at zero and less than one at one. It has a unique fixed point in $(0, 1)$. Starting from the uniform bound one, iterating the map gives successively smaller uniform bounds, including at leaves where the input variance is only σ^2 . Their limit is b_f . Independence of J and X gives $\mathbb{E}(r_J(X)/\sigma)^2 \leq \mathbb{E} \tanh^2(X/2)$.

For the second bound, the normalized input at a nonleaf is $T + S$, with $T \sim N(0, d)$ independent of a centered response sum S and $\mathbb{E}S^2 \leq q\beta$. Its density is bounded below by

$$\mathbb{E} \phi_d(x - S) = \phi_d(x) \mathbb{E} e^{xS/d - S^2/(2d)} \geq e^{-q\beta/(2d)} \phi_d(x),$$

by Jensen. Since $|r(Z, x)| \leq \min(|Z|, |x|)$,

$$\mathbb{E}Y^2 \leq 1 - e^{-q\beta/(2d)} \mathbb{E}(Z^2 - T^2)_+ = 1 - D_d e^{-q\beta/(2d)}.$$

The last integral is evaluated in polar coordinates. At a leaf, $\mathbb{E}Y^2 \leq \mathbb{E} \min(Z^2, G^2) = 1 - 2/\pi < 1/2$. The second fixed-point map sends $[1/2, 1]$ into itself and has derivative less than one, since $0 < D_d \leq D_3 < 1/2$. Its fixed point therefore bounds all depths by induction. This proves (32).

B.2 The sibling integral

For $w > 0$, put $m(u) = \cos(\min(|u|, \pi))$ and define

$$s_b(w) = \begin{cases} \mathbb{E}[m(\sqrt{b/w}G)^{q-1}], & q-1 \text{ odd,} \\ \mathbb{E}[(m(\sqrt{b/w}G)_+)^{q-1}], & q-1 \text{ even.} \end{cases} \quad (33)$$

If S is the sum of $q - 1$ iid symmetric responses with variance at most b , then $\mathbb{E}e^{-S^2/(2w)} \geq s_b(w)$. Indeed, the Gaussian Fourier identity and independence give

$$\mathbb{E}e^{-S^2/(2w)} = \mathbb{E}_G[\mathbb{E}_Y \cos(GY/\sqrt{w})]^{q-1}.$$

For a symmetric Y with second moment at most b , $\mathbb{E} \cos(tY) \geq \cos(\min(|t|\sqrt{b}, \pi))$. To check this inequality when $|t|\sqrt{b} \leq \pi$, consider $g(u) = \cos(t\sqrt{u})$ for $u \geq 0$. It is convex up to $u = \pi^2/t^2$. Its tangent at $u = \mathbb{E}Y^2$ lies below it there; at π^2/t^2 that tangent is at most -1 , and remains a lower bound beyond that point because it is nonincreasing. Taking expectations of the tangent proves the assertion. When $|t|\sqrt{b} > \pi$, the lower bound -1 is automatic. Odd powers preserve this lower bound. For even powers use its positive part, giving (33). Also

$$s_b(w) \geq 1 - \frac{(q-1)b}{2w} > 0 \quad (w \geq d).$$

For odd powers this follows from $m(u) \geq 1 - u^2/2$ and $m(u)^{q-1} \geq 1 - (q-1)u^2/2$; if $u^2/2 > 1$, the right side is at most -1 for odd exponents at least three. The exponent one is direct. The even, clipped case follows from the same inequality on $u^2/2 \leq 1$ and nonnegativity outside it.

B.3 A Gaussian certificate

Choose $\tau \geq d$, $0 < \varepsilon < 1$, and a truncation power $p_0 \in \{1, 2\}$. Using (23), put

$$\begin{aligned} f(z) &= \min\{1, (c(z)/\varepsilon)^{p_0}\}, \quad N = \mathbb{E}[f(Z)^2/c(Z)^2], \\ \Phi(x) &= \mathbb{E}\left[f(Z)\psi(Z, \sqrt{d}G + x)e^{-r(Z, \sqrt{d}G+x)^2/(2\tau)}\right]. \end{aligned} \quad (34)$$

The ratio in N has continuous limit ε^{-2} for $p_0 = 1$ and zero for $p_0 = 2$. Suppose positive coefficients α_j and widths $w_j \geq \tau$ satisfy the *global* inequality

$$\sum_{j=1}^J \alpha_j e^{-x^2/(2w_j)} \leq \Phi(x) \quad \text{for every real } x. \quad (35)$$

Checking a finite grid of x alone does not establish this condition. Define

$$\Lambda = \sum_j \alpha_j s_b(w_j), \quad a = q\Lambda, \quad v = e^{-b/(2\tau)}, \quad \delta = \frac{\log(qN)}{2\log a} - 1.$$

Require $a > 1$ and $\delta > 0$. The conditional path argument leading to (26) now uses

$$\begin{aligned} \mathbb{E}\Phi(S + y) &\geq \sum_j \alpha_j e^{-y^2/(2w_j)} \mathbb{E}[e^{-S^2/(2w_j)} \cosh(Sy/w_j)] \\ &\geq \Lambda e^{-y^2/(2\tau)}. \end{aligned}$$

The last response has auxiliary expectation at least v by (32). Consequently the following proposition follows from the same projection lemma.

Proposition B.1 (Finite-depth and polynomial certificates). *For the constants just defined, every $V \in \mathcal{V}_{2h}$ on a regular tree ball of depth L satisfies*

$$\|V - L_V\|_2 \geq \sigma\sqrt{d}v \max_{\substack{0 \leq n \leq L-1 \\ n \in \mathbb{Z}}} a^{-\delta n} \left(1 - \frac{2h}{dva^n}\right)_+. \quad (36)$$

For a polynomial form, set

$$H = \max \left\{ 1, \frac{2(1+\delta)}{dv\delta} \right\}, \quad c_* = \sigma\sqrt{d}v a^{-\delta} \left(1 - \frac{2}{dvH} \right) H^{-\delta} > 0.$$

If $L \geq 1 + \lceil \log_a(Hh) \rceil$, then $\|\mathbf{V} - L_V\|_2 \geq c_* h^{-\delta}$. All constants in this polynomial statement are fixed as h varies.

Proof. The finite-level statement is (29) with the improved Λ and v . Take $n = \lceil \log_a(Hh) \rceil$ to obtain the polynomial form. Differentiation maximizes $(1 - 2/(dvH))H^{-\delta}$ over $H \geq 1$ at the displayed choice. For the finite-depth maximum, the positive expression increases up to

$$n_* = \frac{\log(2h(1+\delta)/(dv\delta))}{\log a}$$

and decreases thereafter. Clip n_* to $[0, L-1]$ and check the adjacent integers. If the expression is identically zero on the available levels, this procedure still gives its maximum. $\square$

There is a convenient analytic way to obtain (35) without checking infinitely many points:

$$\tau = d + 2/\sigma^2, \quad J = 1, \quad w_1 = \tau, \quad \alpha_1 = \Phi(0). \quad (37)$$

To prove it, let $u = 2/\sigma^2$ and $Q(x) = \mathbb{E}_Z[f(Z)\psi(Z, x)e^{-r(Z, x)^2/(2\tau)}]$. The curvature calculation for (24) says each integrand, after multiplication by $e^{x^2/(2u)}$, is log-convex. Positive mixtures preserve log-convexity by Hölder. Thus $g(x) = e^{x^2/(2u)}Q(x)$ is even and convex. Completing the square gives

$$\Phi(x) = \sqrt{\frac{u}{d+u}} e^{-x^2/(2(d+u))} \mathbb{E}g\left(\frac{u}{d+u}x + G_{du/(d+u)}\right),$$

where $G_a \sim N(0, a)$. The last expectation is even and convex in x , so is minimized at zero. Since $d+u = \tau$, this proves $\Phi(x) \geq \Phi(0)e^{-x^2/(2\tau)}$. The figure data record which analytic or mixture choices were used.

B.4 The two endpoint-mixture cells

Of the 193 populated heatmap cells, 191 use (37). The other two use the following retained computational certificates for *untruncated* Φ , meaning $f = 1$:

d	σ	b used in verification	τ	components (α_j, w_j)
5	4	0.738	5	(0.2680, 5.555), (0.0317, 5.653)
6	8	0.764	6	(0.0725, 6.324), (0.1740, 6.436)

These mixtures were verified using outward ball arithmetic on monotone integration cells, with an analytic tail inequality covering the unbounded part of the real line. The release retains the parameters and the original search-and-verification algorithm. They are computational inputs to the figure, not new symbolic assertions about optimal exponents.

To use them in the projection theorem we still need a finite weight w . Take the linear truncation $f(z) = \min(1, c(z)/\varepsilon)$ and put $u = 2 \operatorname{artanh}(\varepsilon)/\sigma \leq 1$. Let $Q_0(x)$ and $Q_\varepsilon(x)$ be the untruncated and truncated integrands averaged over Z , before convolution with $\sqrt{d}G$. Monotonicity of ψ in $|z|$ and $|r(z, x)| \leq |z|$ give

$$Q_0(x) - Q_\varepsilon(x) \leq \mathbb{P}(|Z| < u)\psi(1, x), \quad Q_0(x) \geq \mathbb{P}(1 \leq |Z| \leq 2)e^{-2/\tau}\psi(1, x).$$

Hence $Q_\varepsilon \geq (1 - \eta_\varepsilon)Q_0$, where

$$\eta_\varepsilon = \frac{\mathbb{P}(|Z| < u)e^{2/\tau}}{\mathbb{P}(1 \leq |Z| \leq 2)}.$$

For $\eta_\varepsilon < 1$, multiplying each tabulated coefficient by $1 - \eta_\varepsilon$ therefore gives a valid truncated certificate. The stored cutoffs are approximately 0.0002890854 at $(5, 4)$ and 0.0007821605 at $(6, 8)$. Evaluating the sibling factors with the smaller fixed-point bound (32), about 0.7374 and 0.7631, gives the retained numerical exponents about 788.27 and 621.69; both lie beyond the displayed color cap. Those decimals are numerical evaluations, distinct from the outward verification of the underlying untruncated mixtures.

C Converting RMS to mean absolute error

A tail estimate supplies the missing bridge between the two error metrics. Let $X = |\mathbf{V} - L_V|$ for $\mathbf{V} \in \mathcal{V}_k$, and suppose a positive number Δ has been proved to satisfy $\|X\|_2 \geq \Delta$. At the root, each pinned contribution is $N(0, 2\sigma^2)$; the $2d$ pinned contributions are independent. The envelope in (20) obeys

$$X \leq W \leq \sum_{i=1}^{2d} |G_i|, \quad G_i \text{ iid } N(0, 2\sigma^2).$$

A union bound therefore gives $\mathbb{P}(W > u) \leq 4de^{-u^2/A^2}$, where $A = 4d\sigma$. This is weaker than (21), but is the normalization used for the retained purple curve. Tail integration yields

$$\mathbb{E}[W^2 \mathbf{1}_{W>u}] \leq 4d(u^2 + A^2)e^{-u^2/A^2}.$$

Splitting X^2 according to whether $X \leq u$ gives $\Delta^2 \leq u\mathbb{E}X + \mathbb{E}[W^2 \mathbf{1}_{W>u}]$, and hence

$$\boxed{\mathbb{E}X \geq \sup_{u>0} \frac{(\Delta^2 - 4d(u^2 + A^2)e^{-u^2/A^2})_+}{u}}. \quad (38)$$

This is a bound on the population mean. It applies to each finite-depth RMS certificate in Proposition B.1; no asymptotic depth substitution is needed to draw the finite-depth figure.

For fixed d, σ and small Δ , choosing u of order $\sqrt{\log(1/\Delta)}$ makes the tail term at most $\Delta^2/2$. Thus an RMS lower bound of order $h^{-\delta}$ gives a mean absolute-error lower bound of order $h^{-2\delta}/\sqrt{\log h}$. The loss in power explains why the purple curve can sit far below an RMS certificate.

D Reading and reproducing the images

The release includes all ten images as PNG files, their editable plotting and diagram sources, retained plotting inputs, and a hash manifest. The figures are reproduced from those retained results. Reproducing the original Monte Carlo and game-solving campaigns is a different operation; their full raw archives and distributed execution infrastructure are not part of this compact release. The local methods notes give the sampling and numerical qualifications needed to interpret the plots.

The three budget heatmaps share axes and a color scale, but their entries have different meanings. The core panel estimates a crossing of a sample mean on 50,000-vertex partially filled trees. The debate panel brackets a population crossing on the same 50,000-vertex ensemble using guarded numerical enclosures and sequential statistical tests; six boundary colors add the stated approximate 1% mean-error tolerance. The ensemble panel evaluates a size-uniform theorem. The lower-exponent panel evaluates a different metric, RMS, with depth increasing logarithmically in the budget.

The three curve figures reuse the same 64 depth-eight boards, allowing a paired empirical comparison of debate and core error. For each board the error is measured against its own full-tree logit, not an infinite-tree limit. The outer shaded bands are approximate pointwise sampling bands; they are not simultaneous confidence bands over every budget.

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